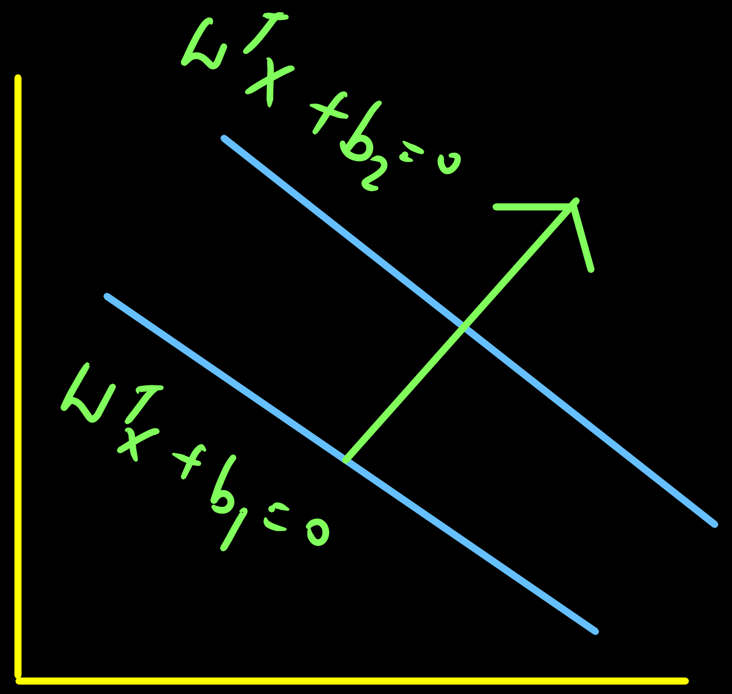
Agenda

1. Distance b/w 2 parallel planes
2. Gain Function
3. Perceptron Algorithm

→ Distance b/w 2 parallel planes

→ What is relation of slopes b/w parallel lines ?



$$y = m_1 x + c_1$$

$$y = m_2 x + c_2$$

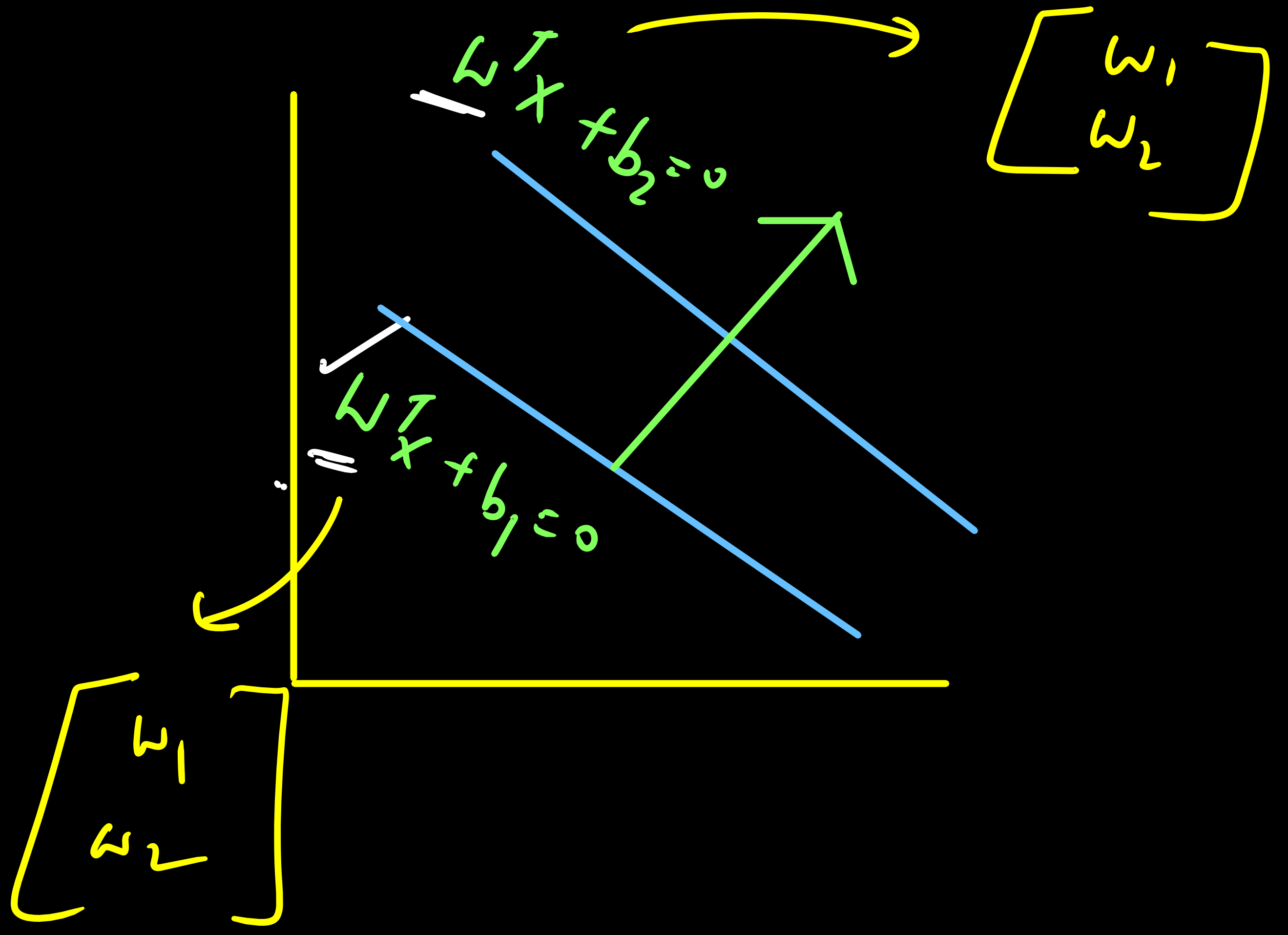
$$\boxed{m_1 = m_2}$$

→ What is relation of slopes b/w parallel lines?

$y = m_1 x + c_1$

$y = m_2 x + c_2$

$\boxed{m_1 = m_2}$



$w_1 x + w_2 y + w_0 = 0$

$w_3 x + w_4 y + w_0 = 0$

$\frac{w_1}{w_3} = \frac{w_2}{w_4}$

$2x + 3y + 2 = 0$

$4x + 6y + 5 = 0$

$2x + 3y + 5/2 = 0$

x_1

↳ Satisfy the equation

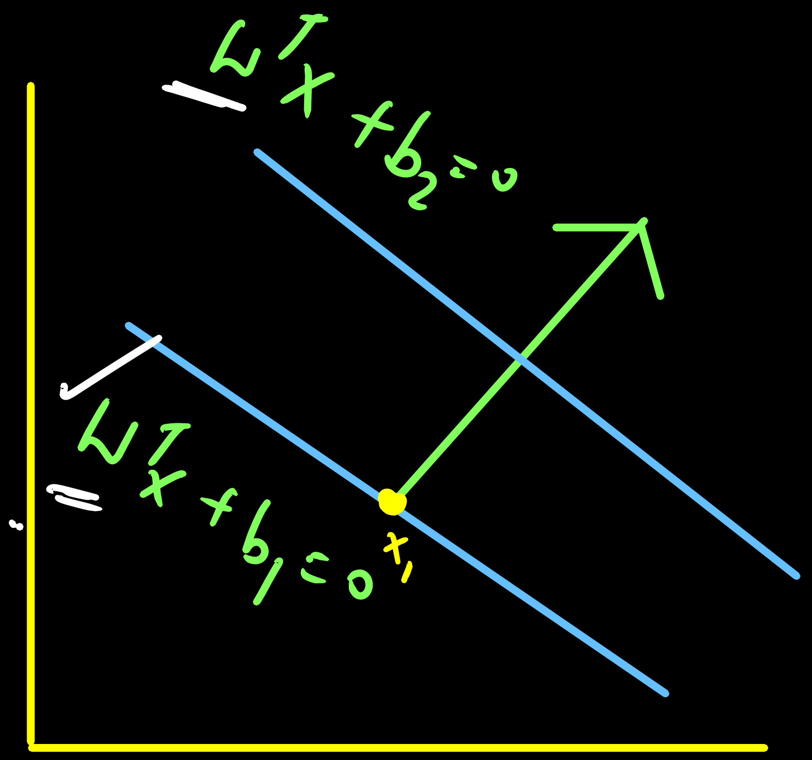
$w^T x_1 + b_1 = 0$

$\longrightarrow w^T x_1 = -b_1$ — 1

$\perp$ dist of x_1 from line $w^T x + b_2 = 0$

$$\frac{w^T x_1 + b_2}{||w||} \quad \text{— 2}$$

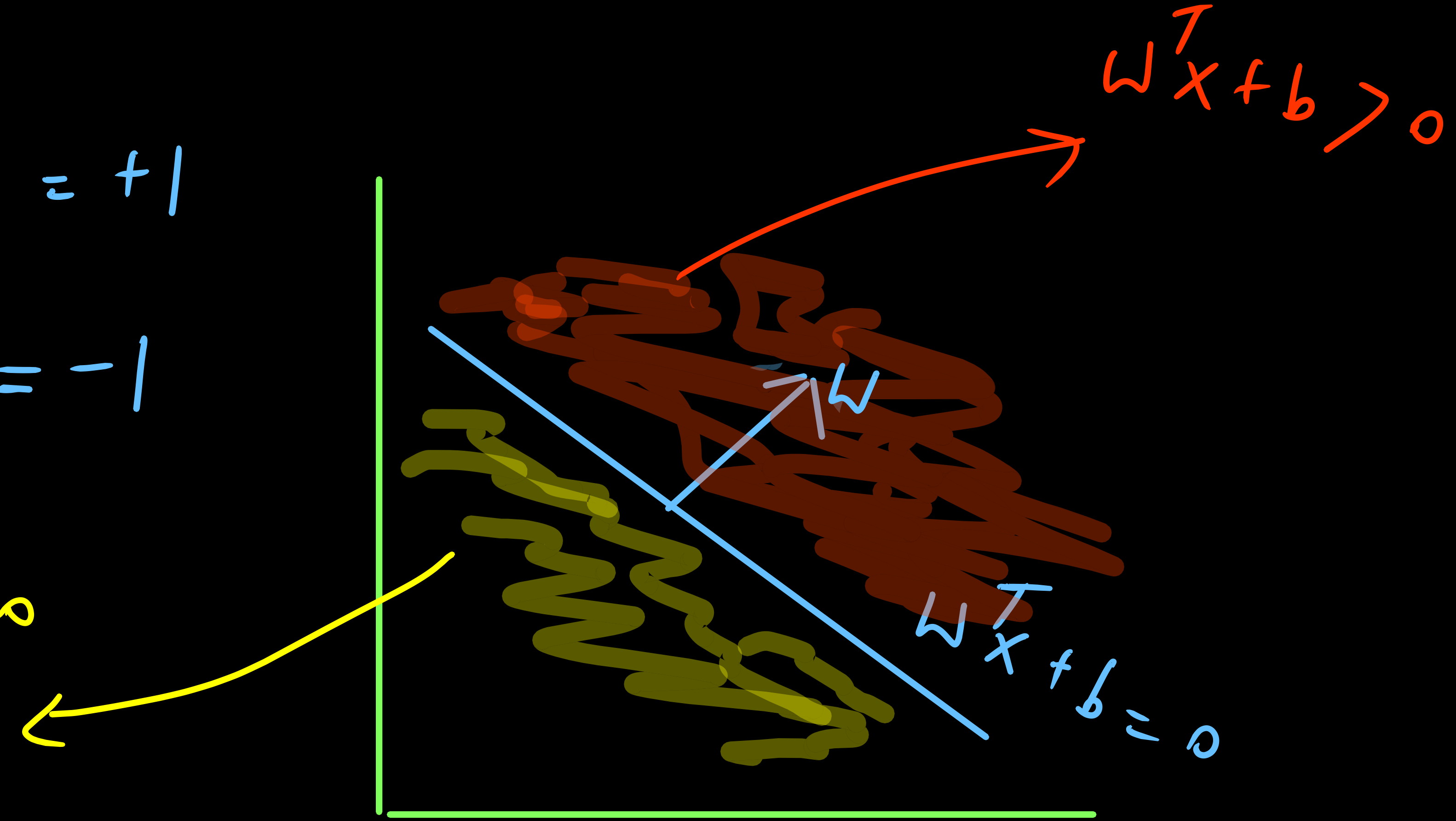
$$\boxed{d = \left| \frac{-b_1 + b_2}{||w||} \right|}$$



Half space

$$w^T x + b > 0 \longrightarrow \hat{y} = +1$$

$$w^T x + b < 0 \longrightarrow \hat{y} = -1$$



Target

X_1	X_2	y
—	—	o (+1)
—	—	o (+1)
—	—	x (−1)
—	—	o
—	—	x
—	—	o
		x

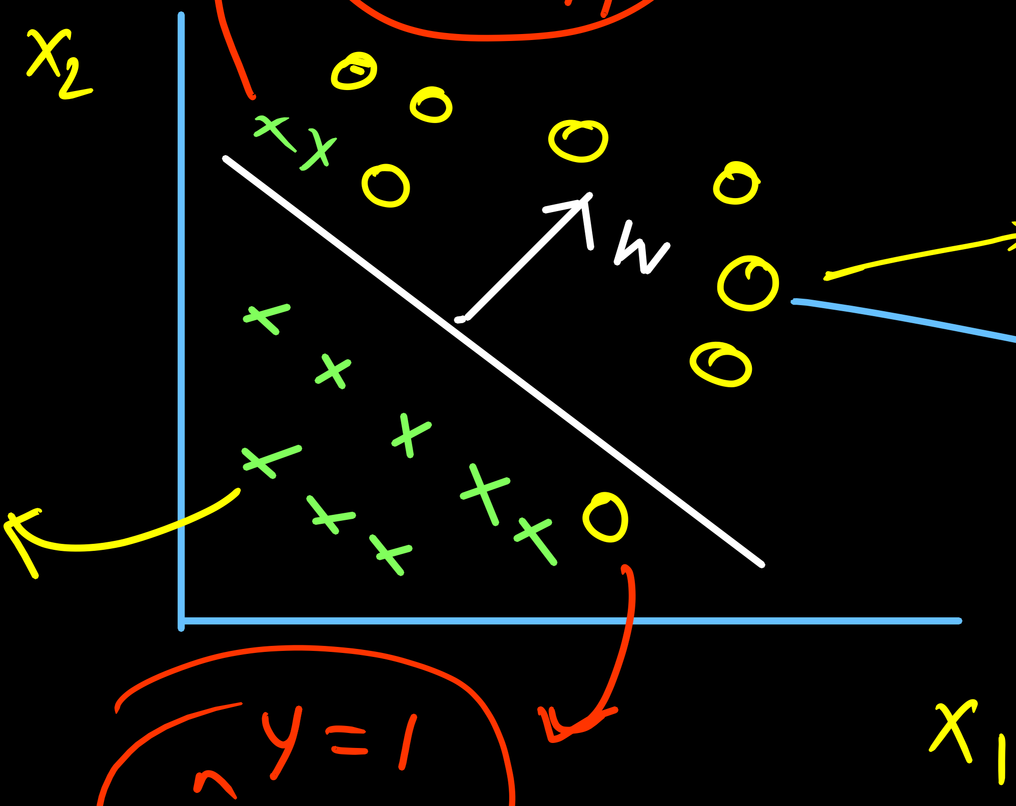
$\hat{y}$

$y^{(1)} = 1, \; \hat{y}^{(1)} =$

o → Dog

x → Cat

$y = -1$
$\hat{y} = +1$



$\hat{y}(i)$

$y = -1$
$\hat{y} = -1$

$Y = +1$

$\hat{y} = +1$

$y = 1$
$\hat{y} = -1$

o →

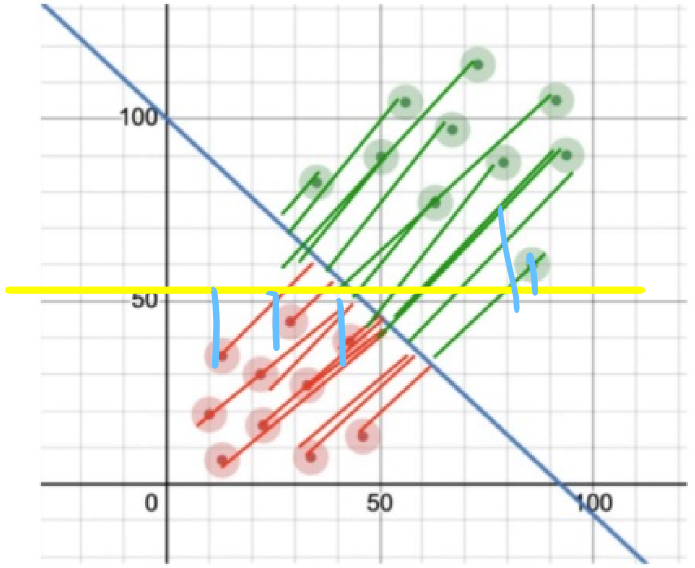
x →

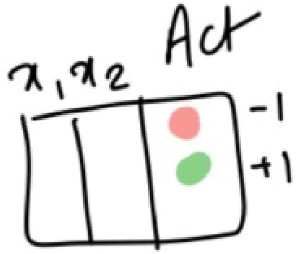
$$w^T x + b > 0 \longrightarrow \hat{y} = +1$$

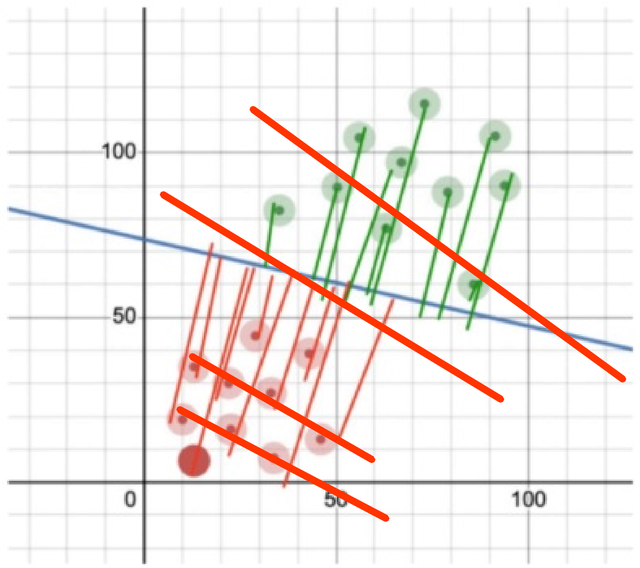
$$w^T x + b < 0 \longrightarrow \hat{y} = -1$$

→ Reduce Misclassification Error

Q) Which line in the figure below is a better comparison?

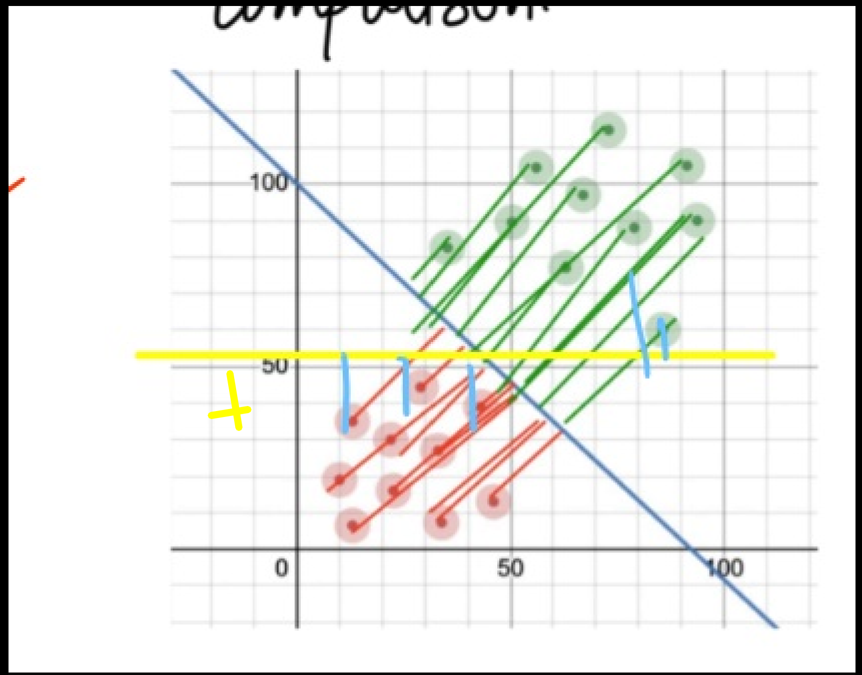






↳ Confidence is higher as distance is more

$$d^{(i)} = \frac{(w^T x^{(i)} + w_0)}{\|w\|}$$

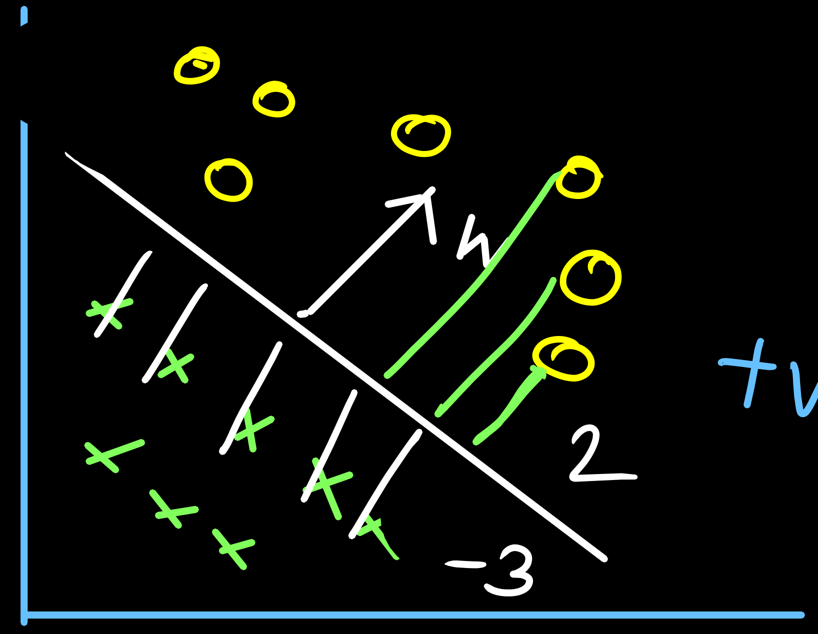


$$G = \sum d^{(i)} = \max \sum \frac{w^T x^{(i)} + w_0}{\|w\|}$$

$o \rightarrow +1$

$x \rightarrow -1$

$$d = \frac{w^T x^{(i)} + b}{\|w\|}$$



$$\sum \frac{w^T x^{(i)} + b}{\|w\|} = \sum +ve \; -ve$$

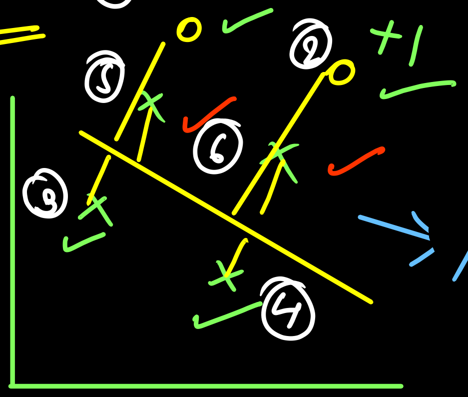
$2 - 3 = -1$

$2 + 3 = 5$

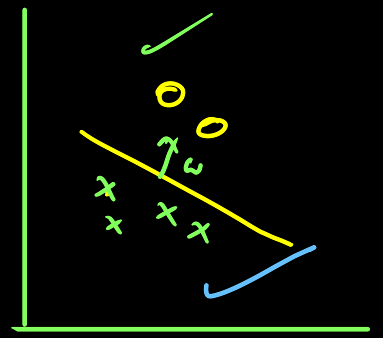
→ $\sum \left| \frac{w^T x^{(i)} + b}{||w||} \right|$ → Cannot be differentiable

→ Square

1



2



o → +1

x → -1

$d_1 \times 1$

$-d_2 \quad -1$

$= d_1 + d_2$

$$\sum \left(\frac{w^T x^{(i)} + b}{||w||} \right)^2$$

$$\frac{y^{(i)} (w^T x^{(i)} + b)}{||w||}$$

1, 2 → +ve × +ve = +ve

3, 4 → −ve × −ve = +ve

Mis-classified point → 5, 6 → −ve × +ve = −ve

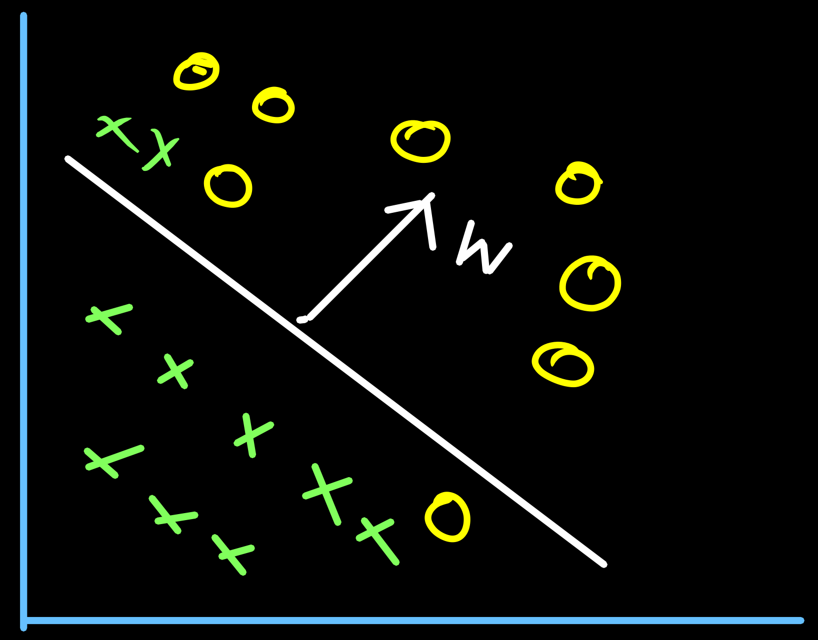
Multiply by sign

$y_i : o \longrightarrow +1$

$y_i : x \longrightarrow -1$

$$G = \max_{w,b} \sum \frac{y^{(i)}(w^T x^{(i)} + b)}{\|w\|}$$

$$L = \min_{w,b} -\sum \frac{y^{(i)}(w^T x^{(i)} + b)}{\|w\|}$$

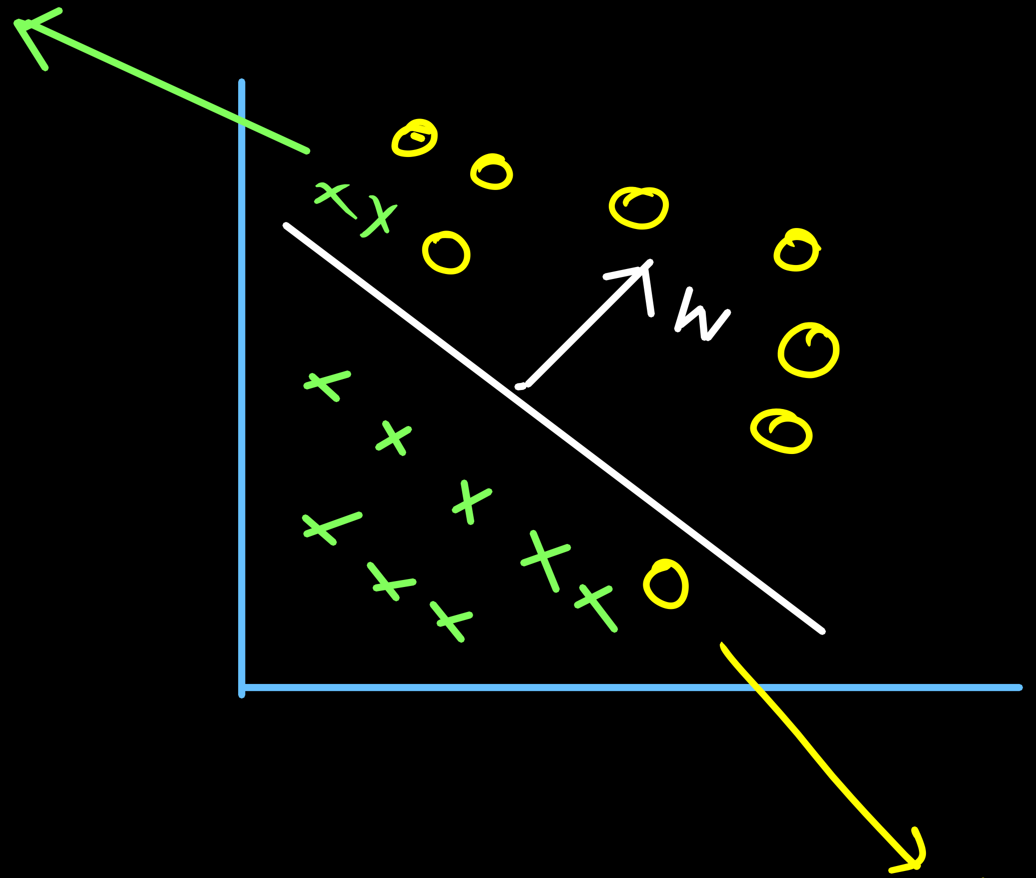


Algo1	Algo 2
Gain = 5	7 ✓
Loss = -5	-7 ✗

Misclassified point

$y_i =$

$\hat{y}_i =$



$y_i =$

$\hat{y}_i =$

Gain Function

$$\frac{y^{(i)}\left(w^T x^{(i)} + b\right)}{\|w\|}$$

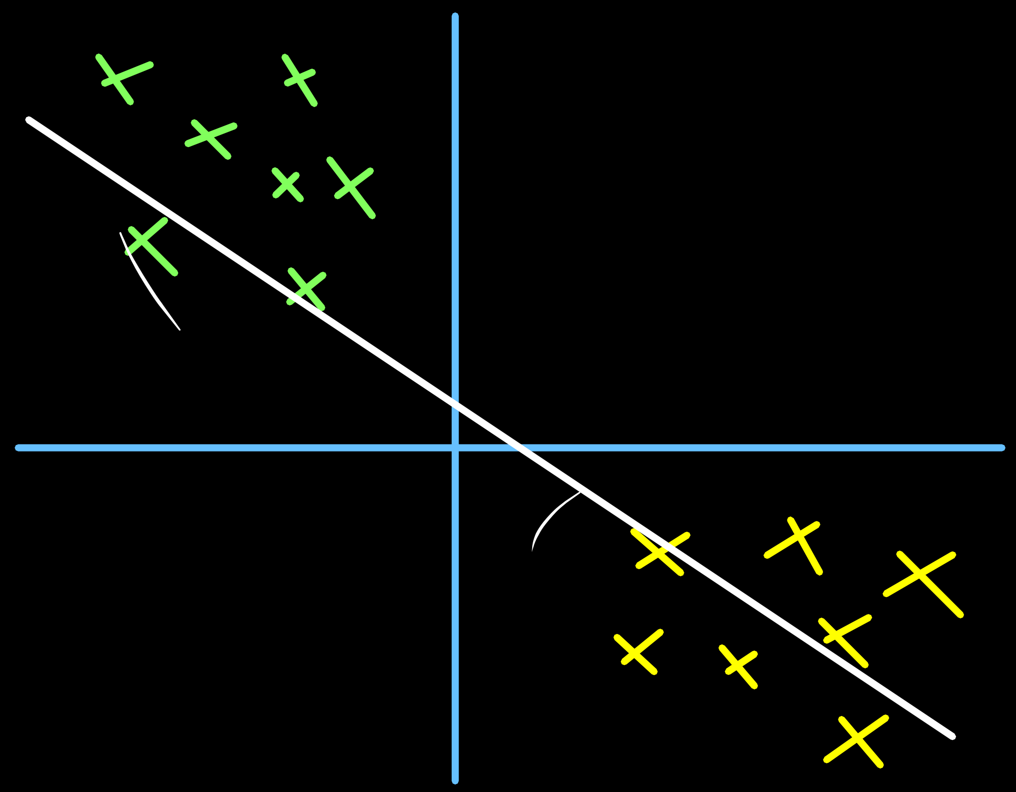
→ Break until 22:23 PM

Perceptron

1. → Start from random hyperplane

2. Look for **misclassified point**

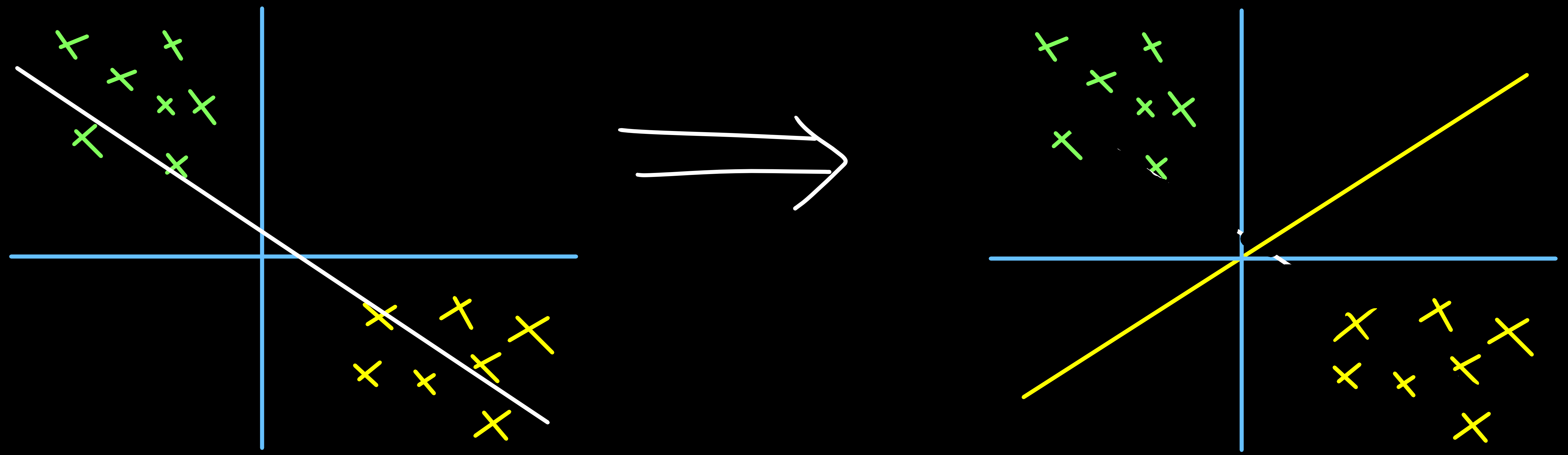
3. Update w to correct missclassification



Target $\rightarrow$ $y^{(i)}$

$\rightarrow$ $\hat{y}^{(i)} = \text{sign}(w^T x^{(i)} + b)$

$\rightarrow$ After training we would like to get this



$$\text{sign}(x) = \begin{cases} 1 & x > 0 \\ -1 & x < 0 \end{cases}$$

Target $\rightarrow$ $y^{(i)}$ — $+1, -1$

$\rightarrow$ $\hat{y}^{(i)} = \text{sign}(w^T x^{(i)} + b)$

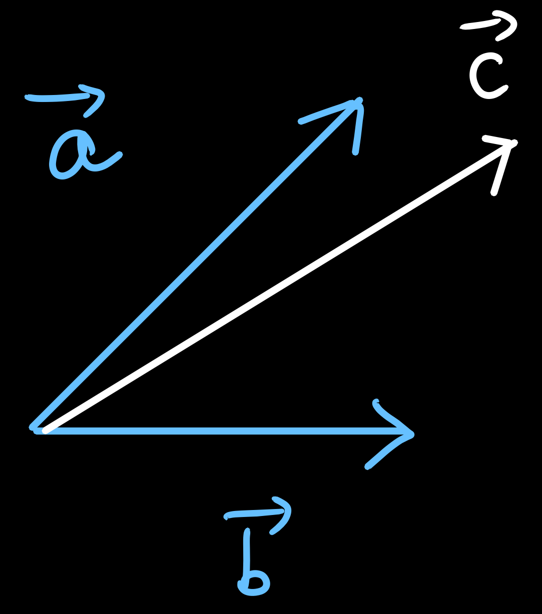
$= +1, -1$

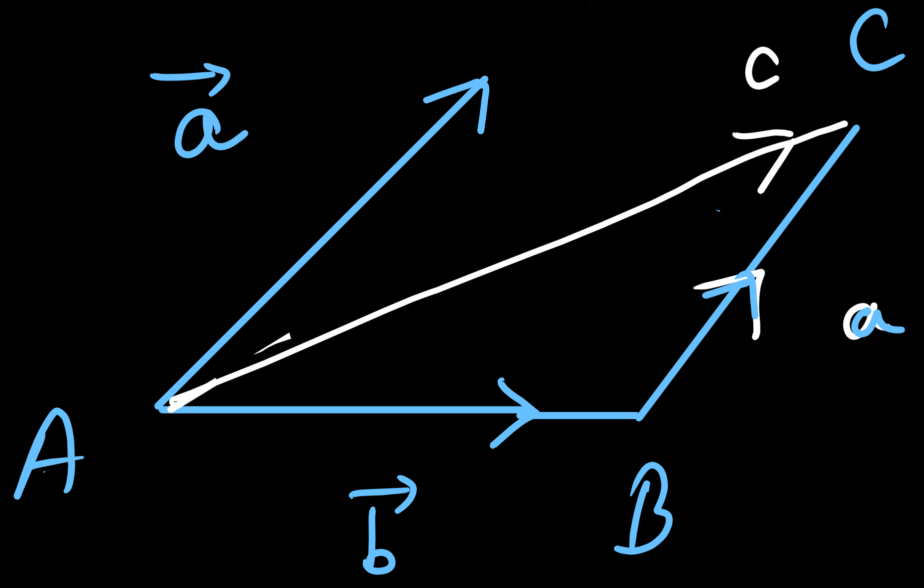
$x^{(i)}$ $y^{(i)}$

	y
$x^{(1)}$	$y^{(1)}$
$x^{(2)}$	$y^{(2)}$

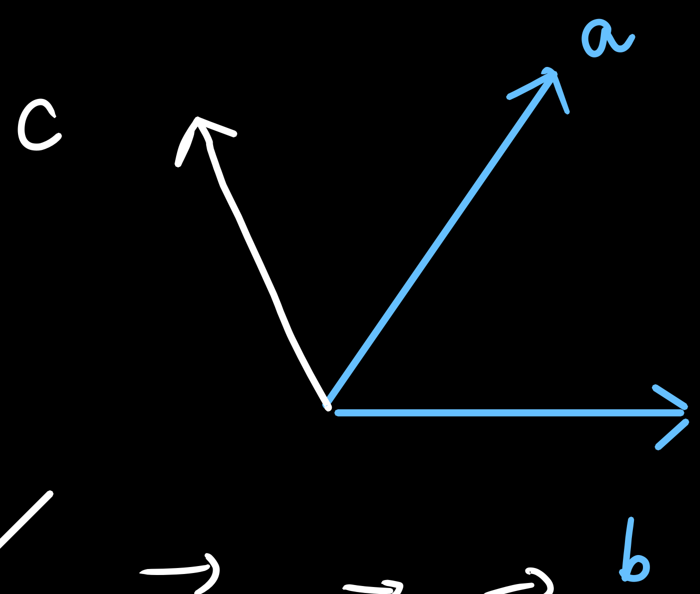
Misclassified

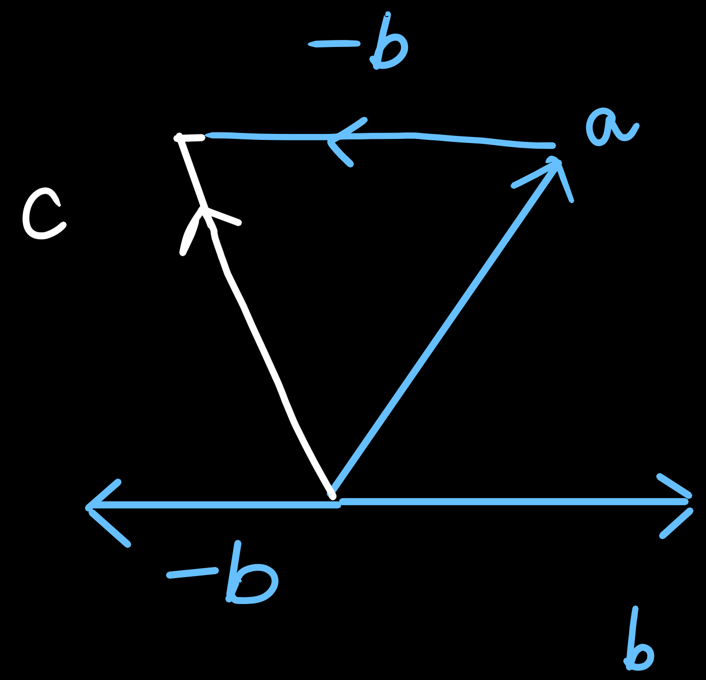
↳ $y^{(i)} \neq \hat{y}^{(i)}$



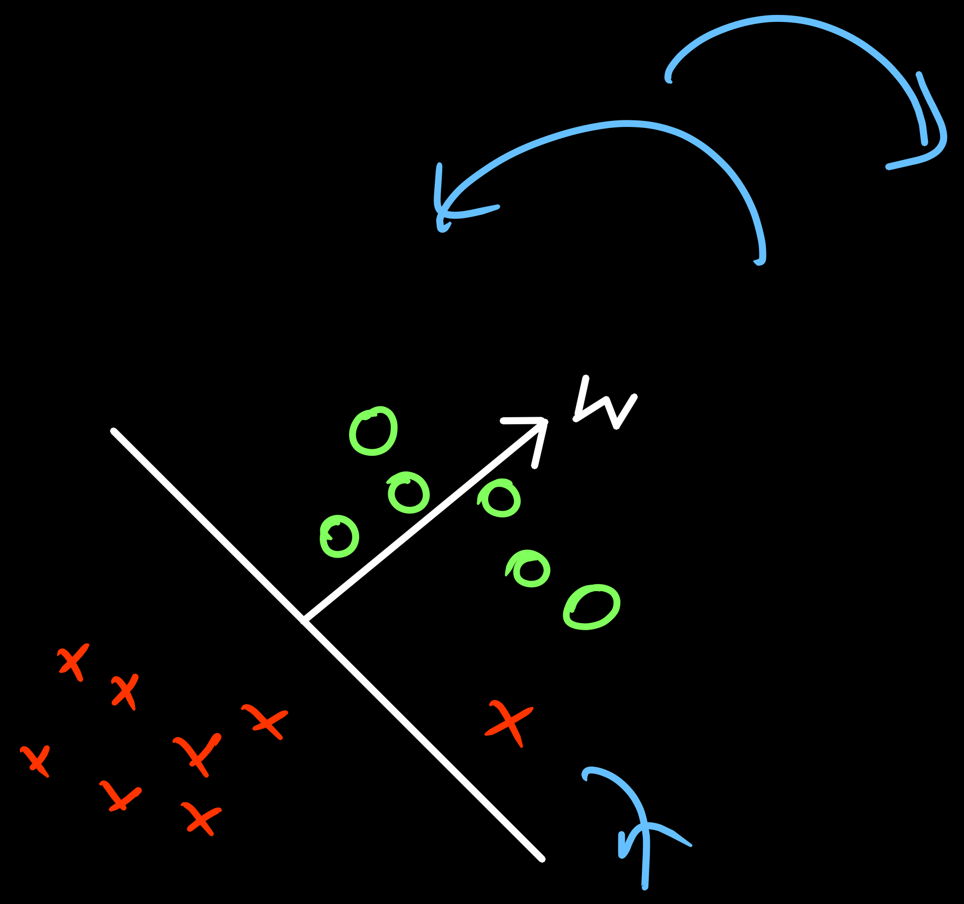


$$\begin{cases} \vec{c} = \vec{a} + \vec{b} \\ AC = AB + BC \end{cases}$$





$$\vec{C} = \vec{a} - \vec{b}$$

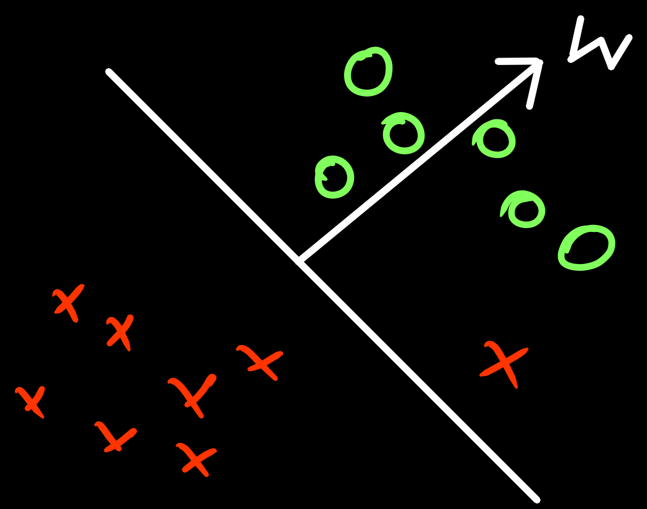


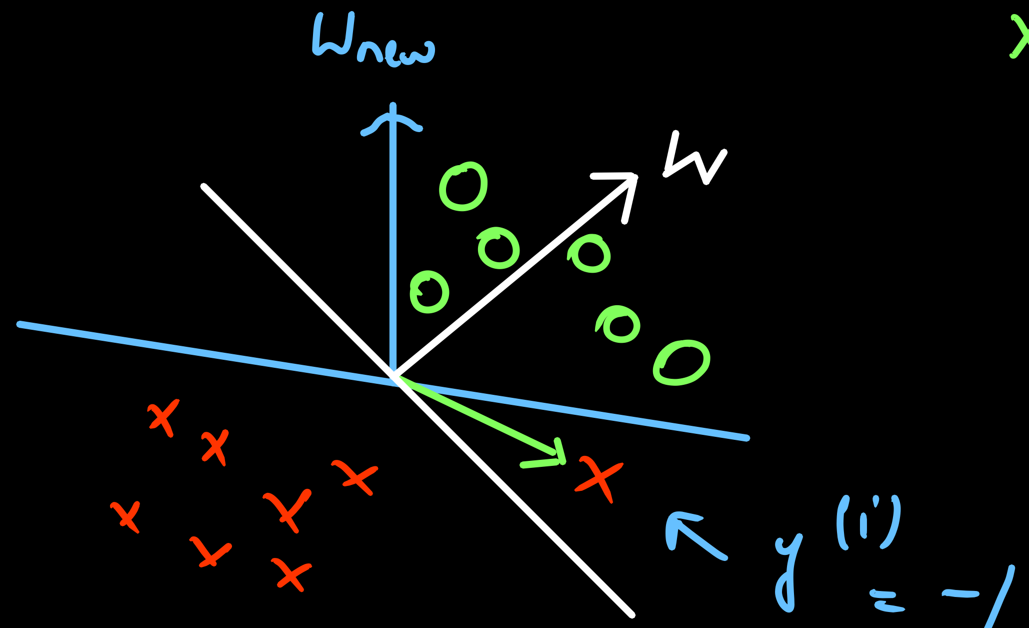
Actual -ve

Predicted +ve

$O \rightarrow y^{(i)} = 1$

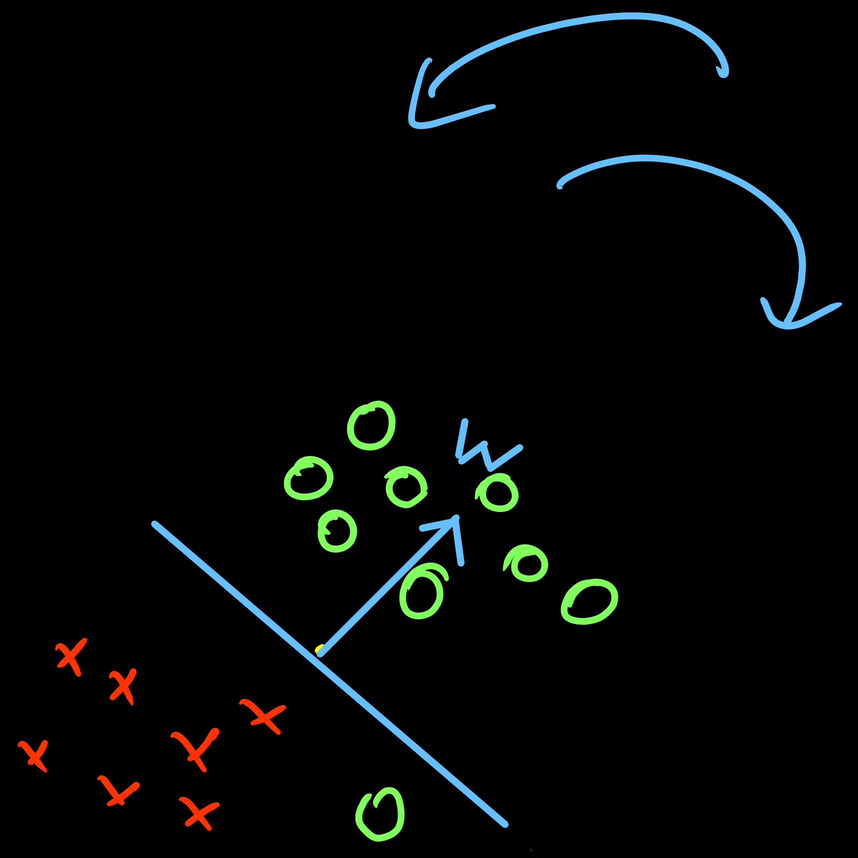
$x \rightarrow y^{(i)} = -1$





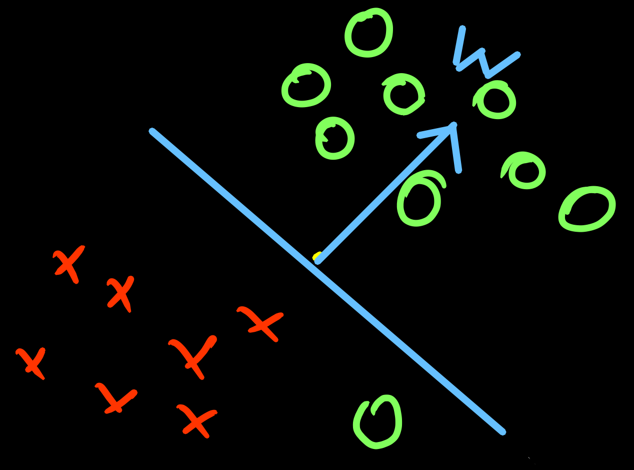
$$W_{new} = W - x^{(i)}$$

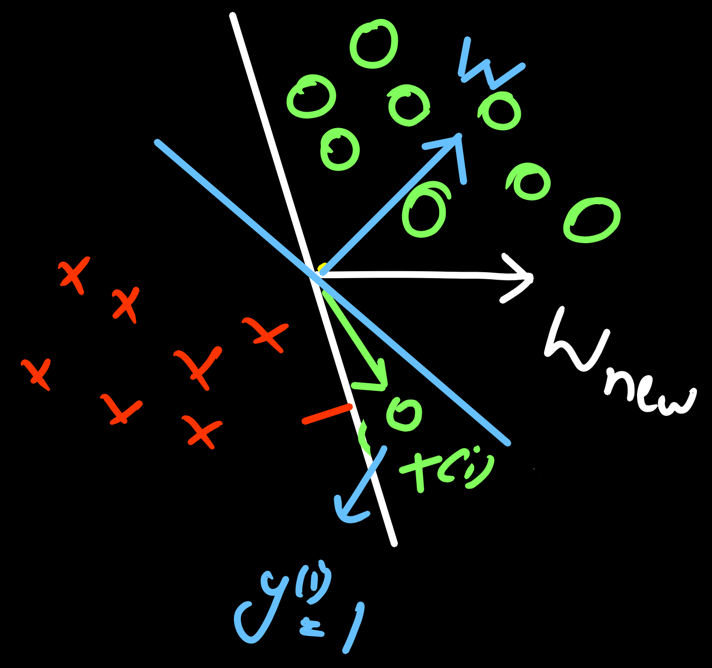
$$= W + \underline{y^{(i)}}\,\underline{x^{(i)}}$$


w

$O \rightarrow y^{(i)} = +1$

$X \rightarrow y^{(i)} = -1$





$$W_{new} = W + X^{(i)}$$

$$= W + y^{(i)} x^{(i)}$$

Missclassified point

$$\text{If}\left(y^{(i)} \neq \text{sign}(w^T x^{(i)} + b) \right):$$

$$W \leftarrow W + y^{(i)} x^{(i)}$$

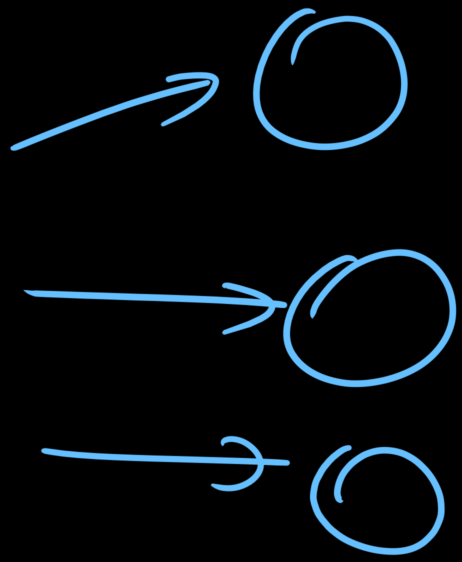
$$b \leftarrow b + y^{(i)}$$

$\{-1, +1\}$

$L = |x|$

$$\frac{\partial L}{\partial x}$$

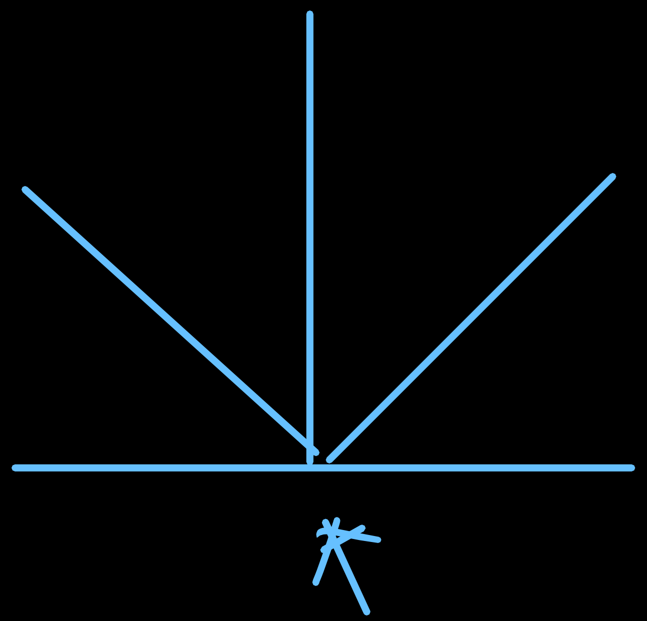
Mutliclass $\rightarrow$ Softmax



A

B

C



$x = 0$